Assignment 3: Working with Amazon Storage Services

Overview

This assignment provides hands-on experience with AWS storage services and data protection patterns. Build a file upload workflow on EC2 backed by EBS and mount a shared filesystem via EFS across instances. Publish a static good webpage in Amazon S3, and answer reflection questions that connect your results to real-world design choices.

Deliverables

Each student must submit the following:

1. **Presentation Slides (10–14 slides maximum)**
    - Title slide with student name(s), course, and assignment title.
    - Clear explanation of each completed task:
        - Task 1 – Working with EBS and shared storage with EFS (**Access Module 7: Storage Lab - 4 Working with EBS from AWS academy**)
        - Task 2 – Create S3 bucket and publish static webpage. (**Access Sandbox: Sandbox Environment from AWS academy**) and follow only the below instruction to access S3 bucket from EC2 instances. Do not follow instruction of Sandbox Environment. Just use Sandbox Environment Lab for S3 bucket.

            1. Create an S3 Bucket
                - Sign in to the AWS Management Console.
                - Navigate to Amazon S3.
                - Click Create bucket.
                - Enter a unique bucket name (e.g., **my-static-website-bucket**).
                - Choose a region close to you.
                - Uncheck “Block all public access” (since a public website requires access).
                - Acknowledge the warning and click Create bucket.
            2. Upload Website Files
                - Open the bucket you just created.
                - Click Upload → Add files.
                - Upload the provided ***index.html***.
                - Confirm and complete the upload.

3. Configure Static Website Hosting
    - In the bucket, go to the Properties tab.
    - Scroll to Static website hosting.
    - Select Enable.
    - Choose Host a static website.
    - For Index document, enter index.html.
    - Save changes.
4. Set Bucket Policy for Public Access
    - Go to the Permissions tab.
    - Under Bucket Policy, add a policy that allows public read access to your objects:

```
{
        "Version": "2012-10-17",
        "Statement": [
                {
                        "Effect": "Allow",
                        "Principal": "*",
                        "Action": "s3:GetObject",
                        "Resource": "arn:aws:s3:::my-static-website-bucket/*"
                }
        ]
}
```

    - Replace **my-static-website-bucket** with your bucket name.
    - Save changes.
5. Test the Website
    a. In the Properties tab, under Static website hosting, copy the Endpoint URL.
    b. Paste it into your browser to see your hosted ***index.html***.

    Remember to delete the S3 bucket after you have done to avoid unnecessary costs.

- Task 3 – Reflection questions

1. How does Amazon EBS provide persistent storage compared to the default storage of an Amazon EC2 instance?
2. What is the importance of snapshots in protecting and recovering data in EBS?
3. What did you learn about managing storage (attach, detach, reuse) after performing the EBS lab?

4. Why must “Block all public access” be disabled when hosting a static website on S3?
5. What is the role of the **index.html** file in static website hosting?
6. Why is a bucket policy required to allow public read access, even after uploading files to the bucket?
7. Why is an AWS IAM Role needed when accessing or managing an S3 bucket securely?

- Screenshots of key steps / results (e.g., df -h with mounts, create S3 buckets, S3 website endpoint etc).
- Reflection question responses (Must be included in slides).
- Lessons learned and challenges encountered.

Presentation Guidelines

- Use professional formatting (legible fonts, consistent colors, organized layout).
- Ensure clarity: avoid overcrowding slides with text; use bullet points and visuals.
- Provide screenshots or videos as evidence of work completed.
- Keep presentations within the time limit (6–9 minutes).
- Be prepared to answer questions about the process and challenges.

1. Task Completion (70 points total)

Each AWS task is evaluated based on correctness, completeness, and evidence (e.g., screenshots and description of output).

2. Presentation Quality (20 points total)

- Clear structure (results, discussion, and question answer)
- Slides visually organized and professional, for example, adding pictures, transitions, animations
- Screenshots included as evidence

3. Professionalism & Delivery (10 points total)

- Confident and clear presentation delivery
- Ability to answer questions from given questions

Submission Instructions

- Submit slides (PowerPoint or PDF) to Moodle by the deadline.